

Revision History

Date	Version	Description	Editor/s	Contributor/s
10/09/21	1.0	Completed	Prashil, Ning, Tong, Shaoxiang, Rafael, John	Prashil, Ning, Tong, Shaoxiang, Rafael, John
18/09/21	1.1	Added section numbering Added table of contents Fixed appendices structure Grammar fixed	Prashil, Ning, Tong, Shaoxiang, Rafael, John	Prashil, Ning, Tong, Shaoxiang, Rafael, John
24/09/21	2.0	Refined project background Improved the purpose of the report Removed project timeline Removed governance, resources, timeframe, schedule control, communications, issue and risk management, QA and recommendation Removed projected scope Refined current scope and initial scope	Prashil, Ning, Tong, Shaoxiang, Rafael, John	Prashil, Ning, Tong, Shaoxiang, Rafael, John
14/11/21	3.0	Added more intended audience Added proposed system Added alternative system Removed initial scope Removed project status Removed scope statement Added figures	Prashil, Ning, Tong, Shaoxiang, Rafael, John	Prashil, Ning, Tong, Shaoxiang, Rafael, John

Consultation Report
for the
Usability and the Functionality of the
BCIS Project Allocation Process

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1. Acronyms and Abbreviations

Acronym/Abbreviation	Definition
AUT	Auckland University of Technology
BCIS	Bachelor of Computer and Information Sciences
CMS	Computer and Mathematical Sciences
ECMS	Engineering, Computer and Mathematical Sciences
EOI	Expression of Interest
L2CC	Layer2 Cloud Connector
HOD	Head of Department
CSSE	Computer Science & Software Engineering

2. Background

Bachelor of Computer and Information Science (BCIS) students work on a real-life industry project as part of their final year in the areas of software development, computer sciences, computational intelligence, IT service science, network and security and analytics. The course prepares AUT students for careers spanning from telecommunications to education and healthcare to finance. In this R&D project students work on the given projects. This experience gives industry or non-profit organisations the option to support the development of young professionals to cover their own demand, and it develops valuable networks and future opportunities for final-year AUT Bachelor of Computer and Information Sciences students.

BCIS students are required to fill and submit an Expression of Interest (EOI) form before the start of each semester. These EOIs contain the projects the students are interested in as well as other information to help with team allocation. The administrator team will then allocate students to appropriate projects based on submitted EOI forms. Allocating students to available projects is one of the key administration activities for a couple of weeks, until the first few weeks of the semester.

However, for the BCIS administration team, R&D paper academic and industry leaders project allocation is a challenging task. The whole process of the project allocation is manual and it becomes more challenging and time-consuming by considering the number of students and available projects. During further investigation, research, and engagement with stakeholders, our project team identified four subsystems and several sub-processes that exist within the current SharePoint system. Among them, some of the work processes are either manual, problematic, or require repetitive tasks. And we realized that the current SharePoint system is an orphan site that is not properly supported by anyone. Therefore, our team decided to propose an alternative system or solution to replace the SharePoint system.

3. Introduction

Our investigation on the processes behind the R&D paper has brought about discoveries that question the functionality and usability of the current system. The processes involved in the R&D paper mainly revolve around AUT's dedicated SharePoint site. From our research, we have found out the current state of SharePoint is not an ideal environment for academics, administrators and students alike. However, the majority of the issues are shared by academics and administrators. The main issues lie within the marking and feedback of assessment items. Other issues such as team management and lack of integration also exist. Ever since the creator of SharePoint for AUT left, the site has been left unsupported with a growing number of problems as time passes. The problems we have identified gave us the opportunity to assist BCIS and R&D by developing a consultation report that analyzes the issues and proposes a recommendation for improvement.

In this report, we provide various diagrams of the current SharePoint system showing the data flow, processes and roles involved. We constructed these diagrams from various interviews and surveys with stakeholders who are part of the system. Furthermore, we analyzed the critical interviews and surveys using a thematic synthesis process to form summarized findings showing the different pain points. We used these pain points and findings to accurately model the current system. Given the problems identified, this report puts forward suggestions for improvement by proposing a recommended system and an alternative system.

4. Purpose

The purpose of this report is to analyze the processes and identify problems with the current SharePoint system which is highly utilized by the R&D paper. With this report, we aim to shed light on the faults of the SharePoint system and provide the rationale on a needed change. We propose a recommendation and alternative for improvement that takes into consideration the issues identified and aims to improve vulnerable processes. This report serves to assist people involved in BCIS and R&D by showing them a model of the current that is easy to understand. We hope that this report can be used by school IT to guide them through which processes need attention and streamlining. We also demonstrate the actions and tools that can be taken to eliminate these issues.

5. Intended Audience

The intended audience for this report are:

- Roopak Sinha, Head of Department for Computer Science and Software Engineering (CSSE)
- Karen Phipps, Interim Director of External Engagement
- Marcella Nielsen, Academic Administrator for Engineering, computer, and mathematical sciences
- Ramesh Lal, Senior Lecturer
- R&D Paper Leader
- R&D Mentors
- R&D Moderators
- School IT

6. Scope

Scope Statement
Project Name: The Usability and Functionality of the BCIS Project Allocation Process Project Sponsor: Ramesh Lal Project Mentor: Tony Clear Date: 7/11/21
Project Justification: We aim to propose a to-be system and alternative for the BCIS project allocation process to assist the R&D paper. We are undertaking this project because of the issues with the current SharePoint system which we have initially discovered from our client. We want to further investigate and potentially discover more issues that would call for improvements that could lead to a major change in the system.
Project Management-related Deliverables: ● Project charter ● Scope statement ● Work breakdown structure ● Project plan ● Status report ● Milestone report ● Stakeholder register ● Stakeholder management strategy ● Project review ● Issue log and risk register ● Feasibility study ● Communications management plan ● Change management plan Product-related Deliverables: 1. Summarized stakeholder findings 2. Thematic synthesis diagrams of critical interviews 3. Current system (as-is system) modelling (swimlane and DFDs) 4. Information bundle 5. Future system (to-be system) modelling 6. Consultation report
Project Success Criteria: ● Successfully modelled the current SharePoint system ● Successfully identified problematic processes ● Successfully analyzed and summarized stakeholder findings ● Proposed recommendation and alternative ● Successfully modelled the future (to-be system) system ● Satisfied client and mentor

7. Current System

7.1 SharePoint Overview

The R&D system starts when the student applies for a programme on the homepage of the AUT website. If the student is eligible, admissions will send a letter of acceptance to that student. After students accept this letter, the admin will check the student's previous education. Arion management will set a compulsory paper for students and send a list of elective papers on the Arion website to let students choose. When Arion determines the student is a BCIS student and will take the R&D paper this year, they will place the R&D paper into that student's schedule for the current semester. After the student submits EOI, CMS Admin will collect all of them and then will start to allocate students into teams and projects based on the student's EOI. After that, CMS Admin will find those students who did not submit their EOI and give them to teams randomly. When the allocation is complete, the paper leader will provide the student guide book, and the CMS Admin will send the project & team confirmation letter to the student.

Once the semester begins, students could contact and meet with the project team, project mentor, project client, and then send a draft project proposal to their mentor. Mentors will send feedback to their teams. Students could refine their project proposals based on this feedback. The student team will present a project proposal to the mentor and moderator when the planning date comes. They will then conduct the mid-project review during the student's proposal presentation. The mentor will mark and confirm the project proposal and provide feedback using the marking form and the moderator will moderate and confirm the project proposal mark and feedback.

After moderators finish their work, CMS Admin will publish the proposals and mid-review progress marks into SharePoint. Then they will email the mid-project progress reviews and proposal grades to students. The moderator will publish the proposals and mid-progress review marks into Arion. Then the moderator and mentor will judge if the project is feasible or not. If their project's first or second proposal presentation is feasible enough, then the mentor and moderator will check if this project needs to be fixed or changed. Students will make necessary changes, then student teams will send a client feedback form to their client, and the client will submit the final client feedback. After that, mentors also submit the final mentor feedback to students.

When students receive the final feedback from the client and mentor, they should submit and present their poster. Then they will work on a portfolio, reflective report, personal logbook, and final product. After finishing those works, the student team will submit them to their client, mentor, and moderator. Then the mentor and moderator will assess and mark submitted components using marking forms. CMS Admin will publish those marks into SharePoint, and they will send the final grades to the students. The moderators are responsible for publishing marks into Arion, and they will also check if there are students with missing marks. After that the whole process ends.

7.2 R&D Assessment Items

BCIS Assessment				
Assessment Item	When	%	Associated learning Goals	Associated Project Goals
Part 1				
Project proposal	Week 6	10	1,5,7,9	Goal identification, feasibility evaluation, Risk evaluation, Scoping and Planning
Mid project progress review	Week 12	10	5,6,7,8	Status and Progress Review, planning and Tracking, Quality Assurance, Risk management
Part 2				
Client feedback	Week 12	5	2,3,4,5	Satisfaction, productivity, professionalism
Supervisor feedback	Ongoing	10	2,4,5,6	Mentoring, Academic Integrity
Poster and Final Product	Poster-Week 13	35	1,2,4,5	Goal and result communication
Reflective Report	Rest-Week 14	10	5	Reflective Practice, Process And Product Evaluation
Portfolio		20	1,2,3,6,7,8	Activity and Process Evidence Evaluation, Product

7.3 R&D Roles

Role	Description
BCIS Student	The student enrolled into the BCIS degree taking on an R&D project.
CMS Administrator	The CMS administrator handles looking for potential clients for projects and assigning students into teams based on their EOIs. as well as assigning the mentors and moderators for each project
R&D Mentor	The mentor is an academic who is assigned to a project and a team of students and their responsibility is to provide guidance, motivation, and role modeling
R&D Moderator	The moderator is an academic who is responsible for reviewing the grades and feedback with the mentor to ensure the grades are correctly given and unbiased. The moderator is responsible for marking most assessment items
R&D Paper Leader	The paper leader manages the information given to students about the paper and provides guidance on how to successfully complete their project throughout the year. The paper leader holds lectures every week containing various useful information
R&D Project Client	The project client is either internal or external. They have reached out to AUT to provide a project. The client provides the project expectations and outcomes. All R&D projects must have a client.
Admissions	The admissions are responsible for accepting applicants to AUT
Arion Management	The Arion management is responsible for handling the students' grades which are given out after each critical assessment item.
School IT Support	The school IT support is responsible for supporting the school's IT needs.

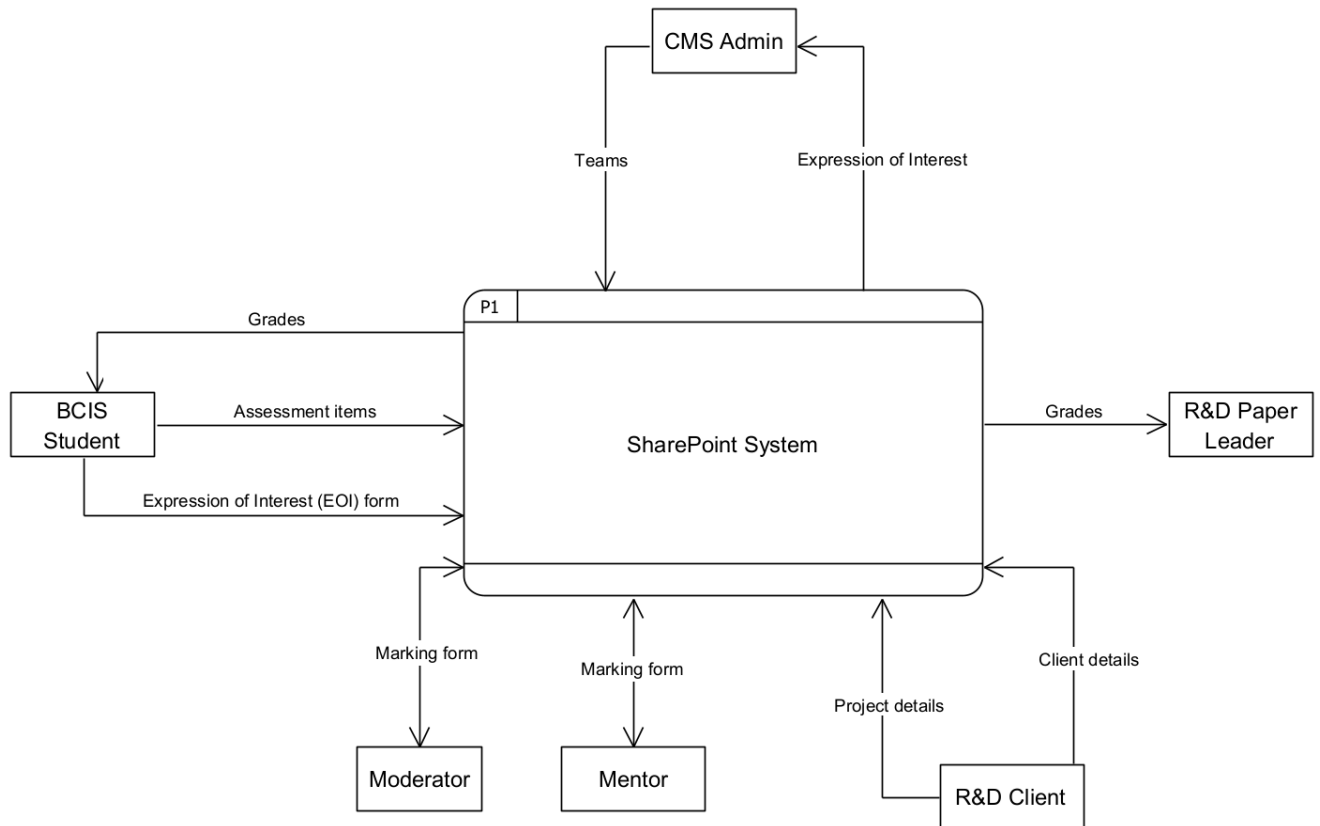
7.4 Pain Points

Role	Pain Points
BCIS Student	• Too little or too many members • Project brief (background, objectives etc.) is insufficient • Require earlier notice of the newly formed team
CMS Admin	• Allocation of teams is time-consuming and stressful • Large amounts of student EOIs being printed • The time frame between receiving EOIs and forming teams is too short (the team have approximately 10 days) • Students requesting for a team change • Academics (mentors and moderators) not reading the guidebook because of their busy schedule • Previous BCIS students appearing in SharePoint site when allocating new students to teams and projects • Manually search for student GPA using Arion • Duplicate projects appearing in SharePoint • Students already in a team are still able to be allocated into another in SharePoint • Marking forms not coming through because both moderator and mentor need to confirm
R&D Paper Leader	• Needing to ring up other academics and/or CMS admin to retrieve missing marks • IT support is non-existent • No one to report bugs and issues to • The usability is a major issue of the SharePoint site
Moderators	• Missing marks (SharePoint not processing moderator and mentor confirmation of marking forms). • Paper leader needing to ring up other academics and/or CMS admin to retrieve missing marks • Repeated teams showing up (already marked and moderated) • Usability is the main issue of SharePoint • SharePoint is not only for BCIS, all irrelevant parts should be hidden • Arion is not connected to SharePoint • Manually extracting marks from SharePoint to Arion (double-entry) • SharePoint is lacking IT Support

Mentor	• Submission of marking forms can create multiple versions (SharePoint registering as multiple submissions) • Allocating a student into already created team does not update the team but creates another one with the new student • Marking forms are not saving • No ability to delete teams and students from teams • Duplicate teams and students appearing on SharePoint
HOD of CSSE	• The guidebook is not professionally written • Academics have a busy schedule • They require a guidebook that is worth their investment of time • SharePoint site is not supported ever since the creator left
Creator of SharePoint for AUT	• Duplicate student from two or more projects • List of unallocated students are not updated automatically • Student did not receive any email notifications for submission • Student submission for documentation and/or product not showing for mentor or moderator

7.5 Data Flow

Fig 1. Context Diagram of the SharePoint System



The context diagram shows the main data points that flow through the SharePoint system along with its origin and destination. Next, we exploded the context diagrams into three top DFDs further exploring the processes and the flow of data.

Fig 2. Top Level DFD of Client and Project Management System

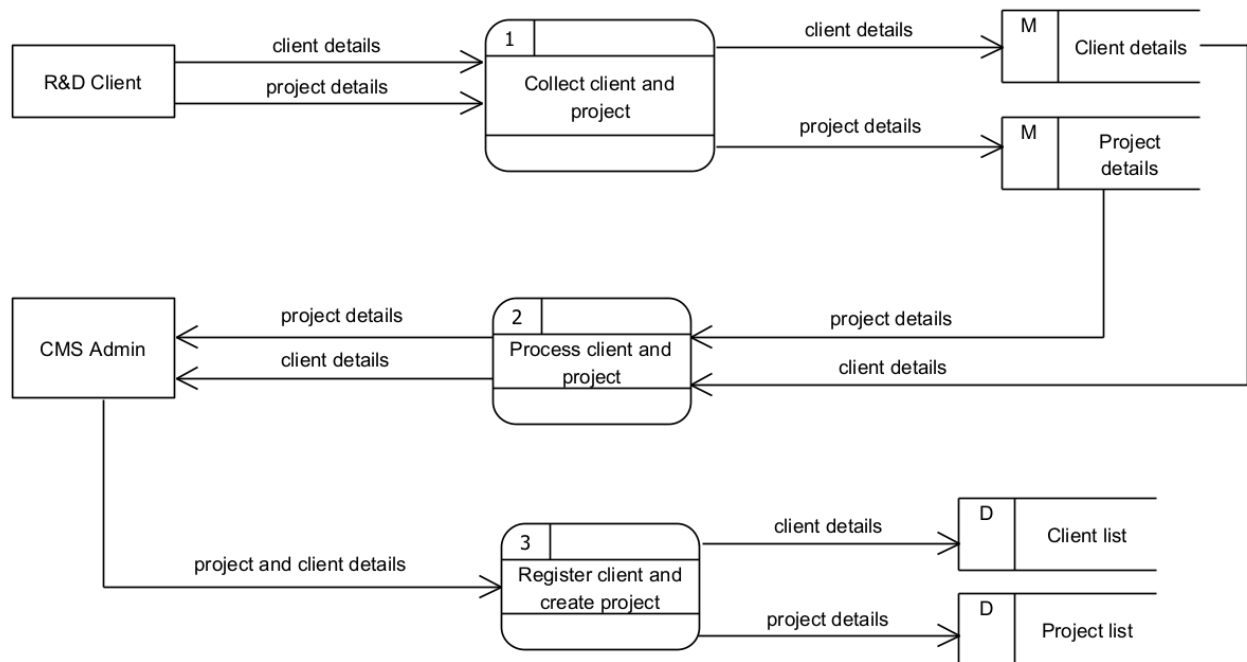


Figure 2 describes the process of retrieving clients for R&D projects which is done by the CMS admin. The client will provide their details and the project they wish to offer to AUT. The CMS admin will process the client and project details. Once processing is done, client details will be stored and the project will be created and made available in the list of projects in SharePoint.

Fig 3. Top Level DFD of Student and Project Allocation System

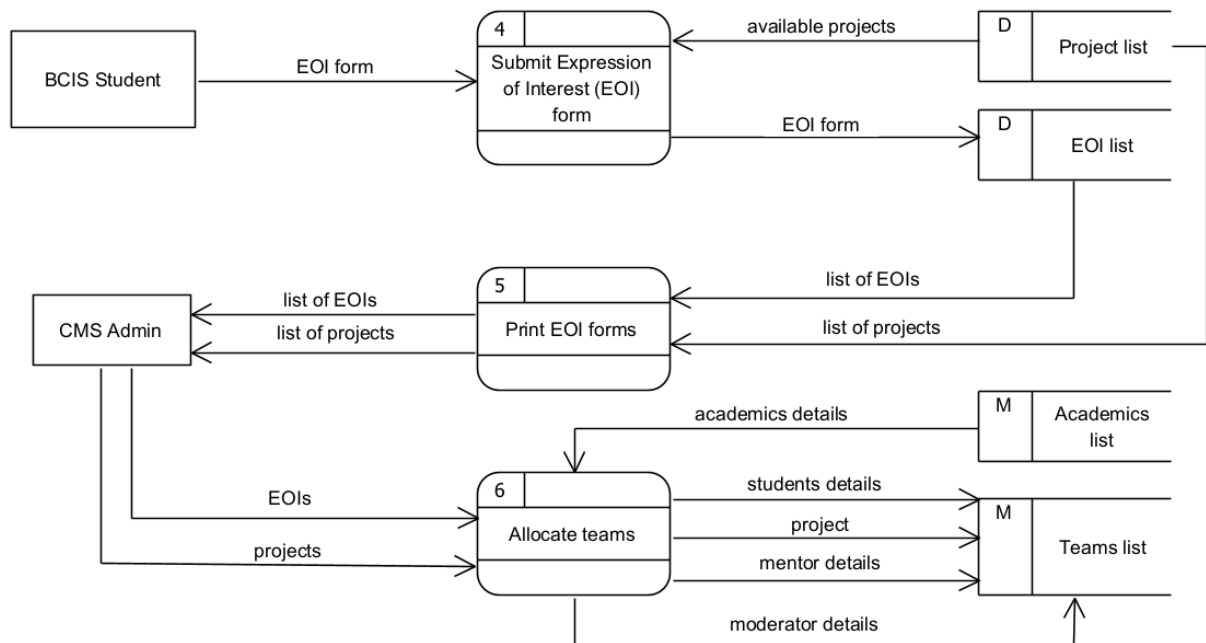


Figure 3 describes allocating students to available projects is one of the key administration activities. A student must submit an Expression of Interest form through SharePoint (BCIS Projects Students Portal). During allocation, the CMS admin will require all EOI forms and projects list to be printed out. The CMS admin will perform a mail merge and export the EOI forms and projects list as an excel file. The EOI forms can be managed in SharePoint (Manage Expression of Interest Requests). All students are allocated by analyzing their EOI forms and matching them to projects along with one mentor and moderator for each team. Students can only be assigned to one project, but mentors and moderators can be involved in one or more projects. The CMS admin will input the teams into SharePoint (Assignment of Mentors, Moderator and Students into Project Teams) after manual allocation. SharePoint also has a workflow that can automatically email the students their team information.

Fig 4. Top Level DFD of Student Marking and Feedback System

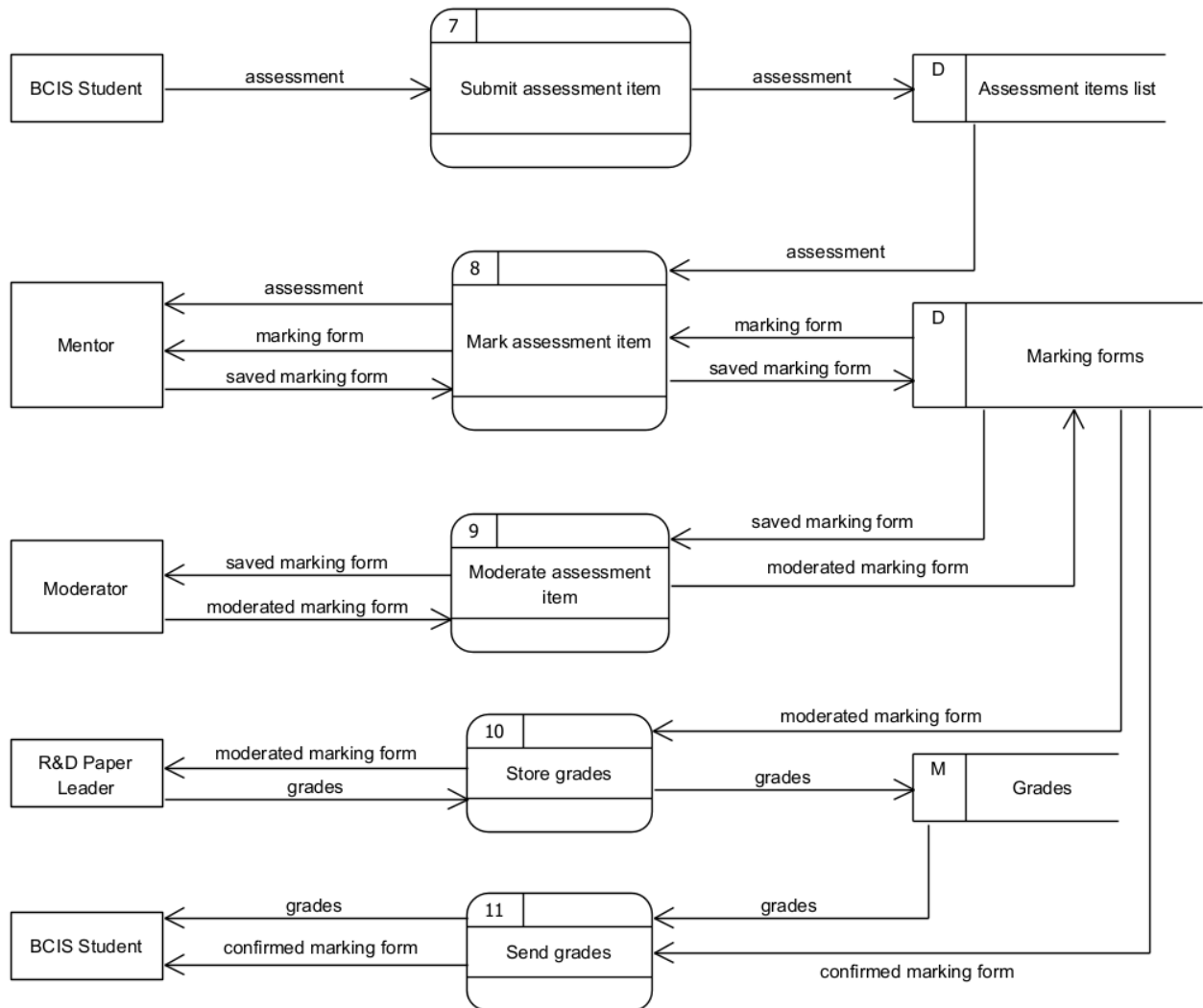


Figure 4 describes the submission and marking of R&D components. There are different ways for submitting R&D components which can be seen as an issue for both students and academics. For instance, the proposal and poster is submitted by email, the reflective report is submitted through Blackboard and the remaining (portfolio, worklog, final product) is submitted through SharePoint (BCIS Projects Student Portal). Each component has a dedicated marking form that mentors and moderators must use to assess. The marking forms are automatically created by Sharepoint. The marking and moderation are performed on the “Student Marking and Feedback Management” part of SharePoint. This system is also used to automatically send grades and feedback to students once the marking forms have been confirmed by the mentor and moderator. Lastly, the R&D paper leader will manually grab the grades from SharePoint and input them to Arion for each student.

7.6 Swimlane Diagrams

The diagrams below are an overview which describe the main processes involved in the SharePoint system for an R&D student to complete their one-year R&D paper. For a more comprehensive understanding of the processes and pain points refer to Appendix {C, D, F}.

Key :

- ❖ **Red Text:** Underlying issues and improvements can be made
- ❖ **Orange Text:** No issues but improvements can be made

Fig 5. Enrolment Overview

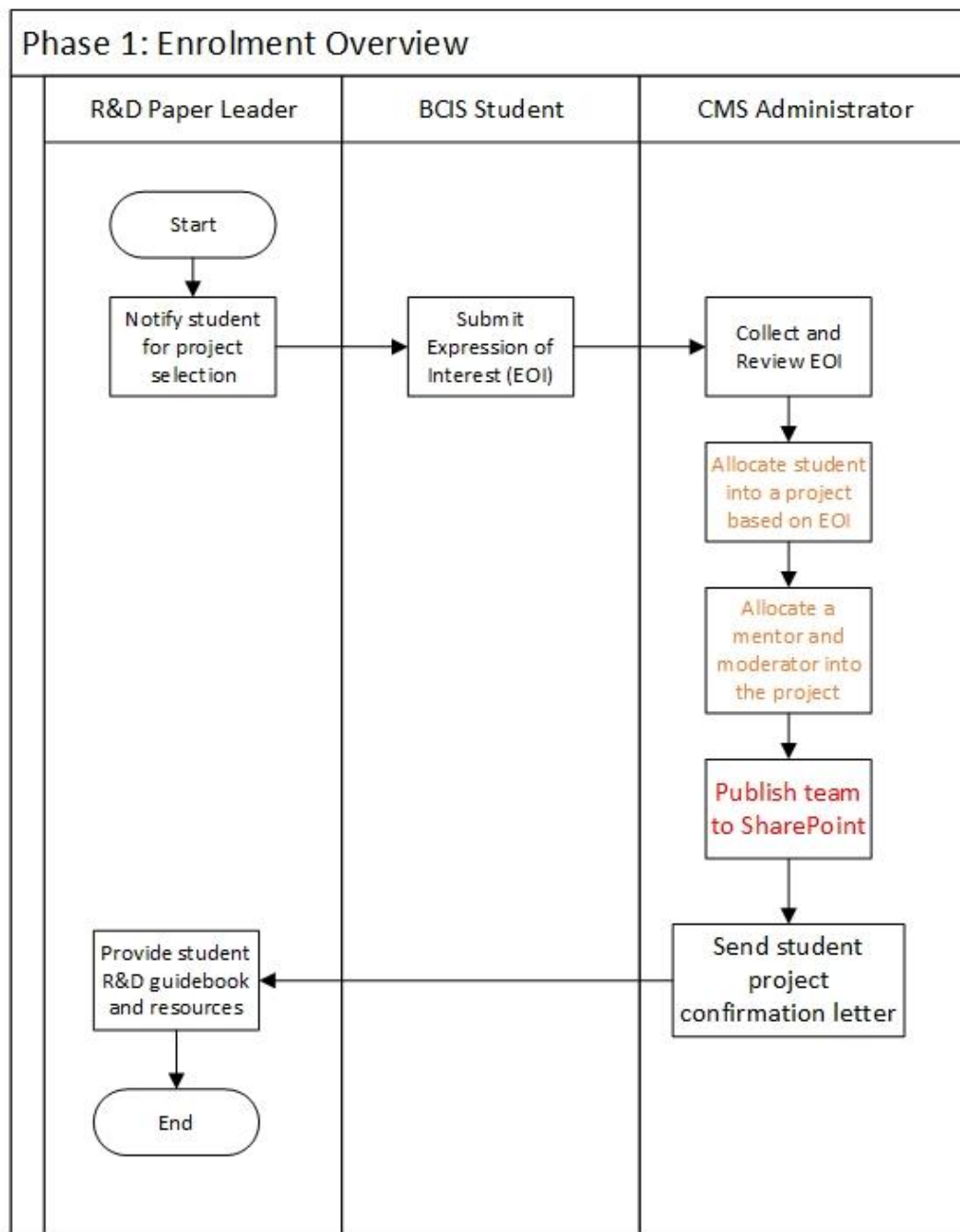


Figure 5 describes the enrolment phase. The process “Allocate students into a project based on EOI” and “Allocate a mentor and moderator into the project” is manual and time-consuming. This process requires a large amount of paper from Expression of Interest (EOI) forms and projects list. The CMS admins must be knowledgeable about the academics to make the decision of which academic will be a mentor or moderator to which project. CMS admins also manually find students' GPA in Arion. The process “Publish team to SharePoint” contains issues such as inability to update and delete a team half way through a project. Instead of updating the team, SharePoint will create a brand new team whilst keeping the previous team. Past BCIS students are also appearing on the list of available students to allocate.

Fig 6. R&D Part 1 Overview

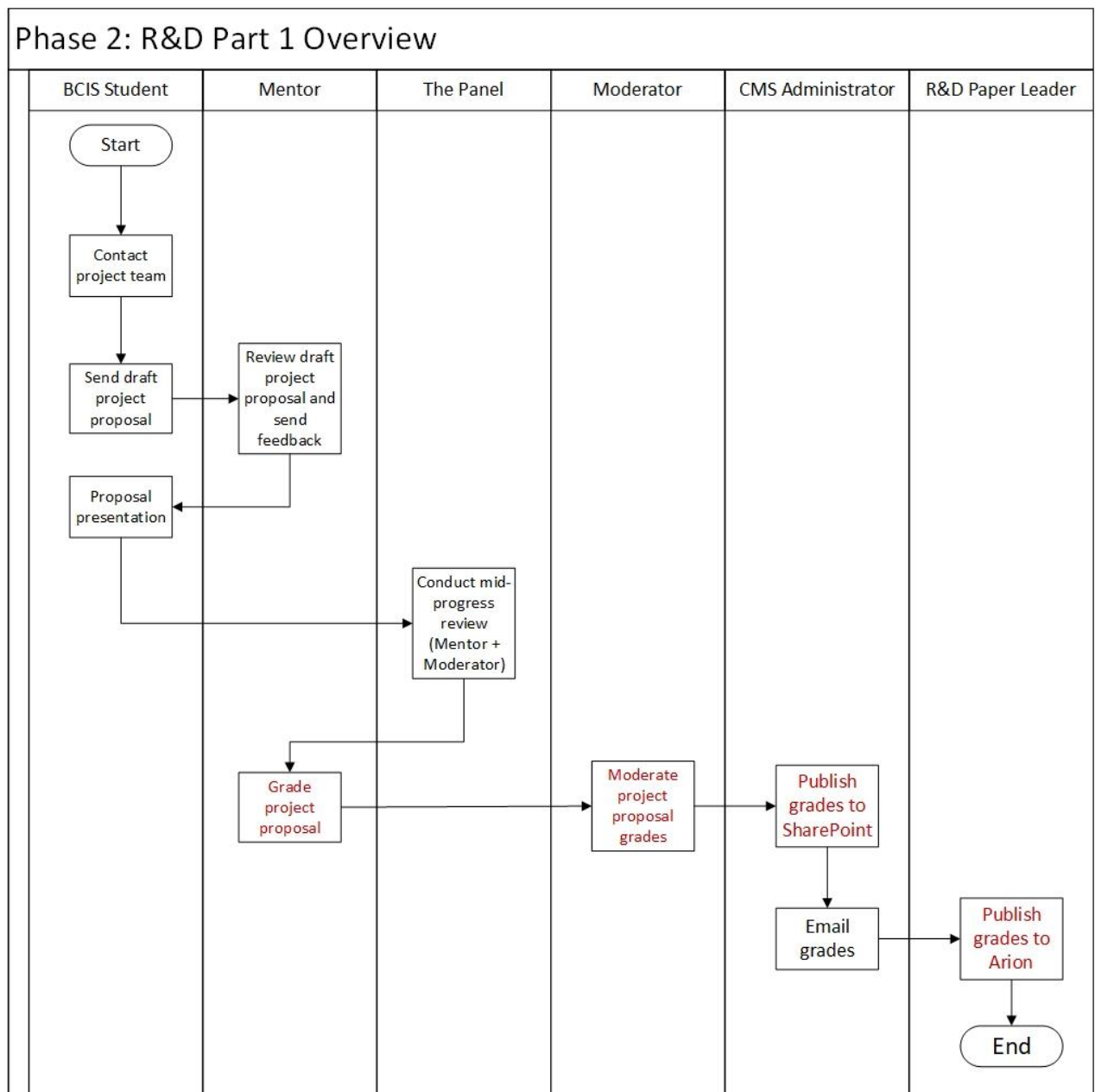


Figure 6 describes the R&D part 1 phase. The process “Grade project proposal” has issues such as marking forms not saving and if duplicate teams exist, mentors may submit a marking form for the wrong team. Another major issue is the guidebook provided which is not well understood by all academics. The process “Moderate project proposal grades” can create duplicate marking forms caused by accidental double submission from the mentor. There is also no way of deleting these marking forms. In some cases, moderation has been completed but is not recognized by SharePoint. This leads to issues in the processes “Publish grades to SharePoint” and “Publish grades to Arion”, where marks are not coming through because both the moderator and mentor need to confirm the marking form. There is also an occurrence of double-entry when inputting grades to Arion and SharePoint.

Fig 7. R&D Part 2 Overview

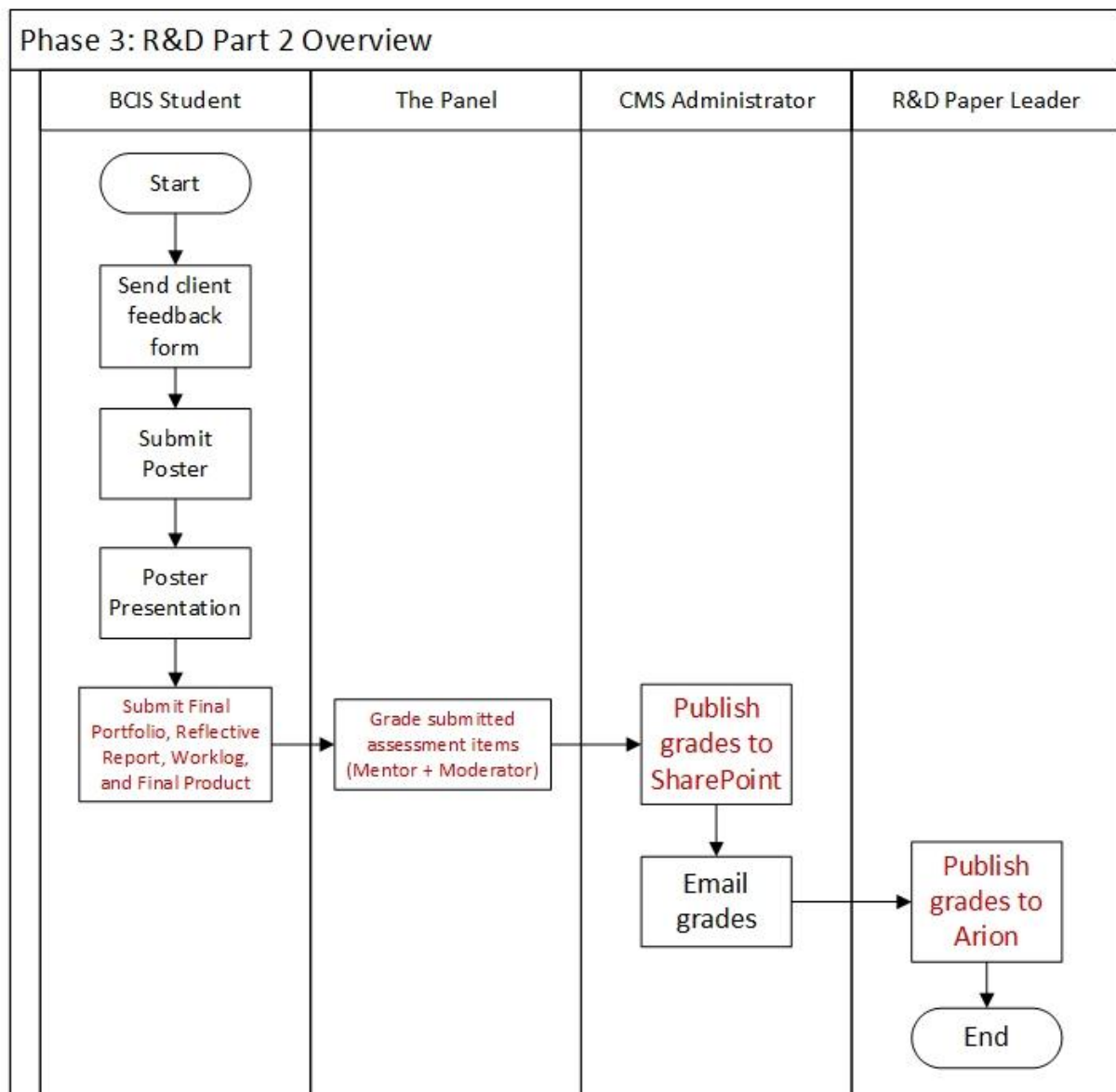


Figure 7 describes the R&D part 2 phase. The process “Submit Final portfolio, Reflective Report, Worklog and Final product” has a bug where student submissions are not displayed for the mentor and/or moderator. The mentor and moderator are also not receiving email notification of students’ submission. These issues only occur when submitting through the SharePoint site. We have not identified any issues with submission through email or BlackBoard. The process “Publish grades to SharePoint” and “Publish grades to Arion” contain the same issues as listed under Figure 6.

8. Proposed System

8.1 Description

The system we are proposing will be a combination of Canvas and SharePoint but in this case SharePoint will have a minor role compared to its current role. In this proposed system SharePoint will only be used for the EOI process and project management, whereas Canvas will take up the role of submissions, grading, allocation and management of the teams. In this system, to grade the students' assessments it will be done by using one of Canvas' tools called the rubrics. To allocate and manage the teams, the groups function will be used to execute this process. With this system, it means that the allocation of the teams will still be manual. Since SharePoint will be handling the EOI process, Microsoft Excel will still be used to export the projects and their relevant details into SharePoint so that the students can determine which projects they like. The project management process will still be handled the same as it is in the current system.

8.2 Modelling

Fig 8. Enrolment Proposed System Overview

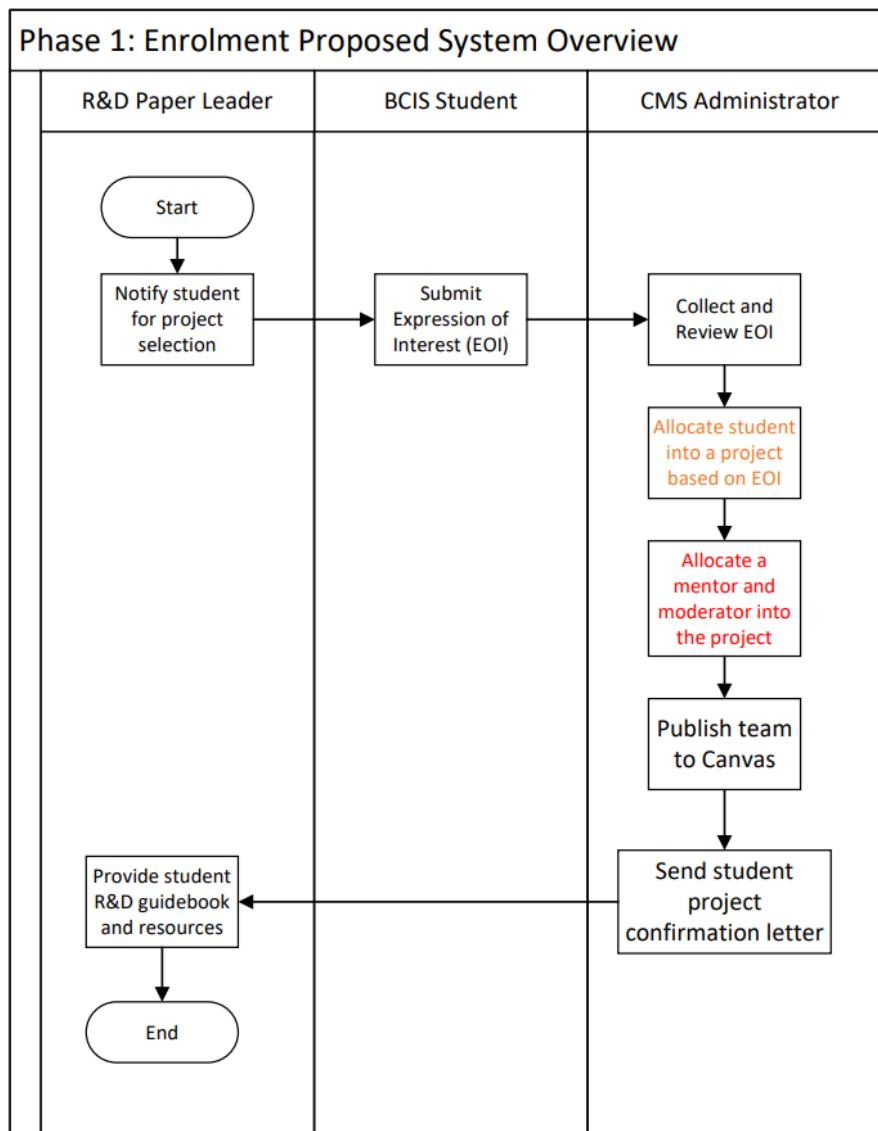


Figure 8 describes our proposed enrolment phase. For the most part it has not changed, the only change that was made was the publishing of the team process. For our proposed system the teams will be published to Canvas rather than SharePoint.

Fig 9. R&D Part 1 Proposed System Overview

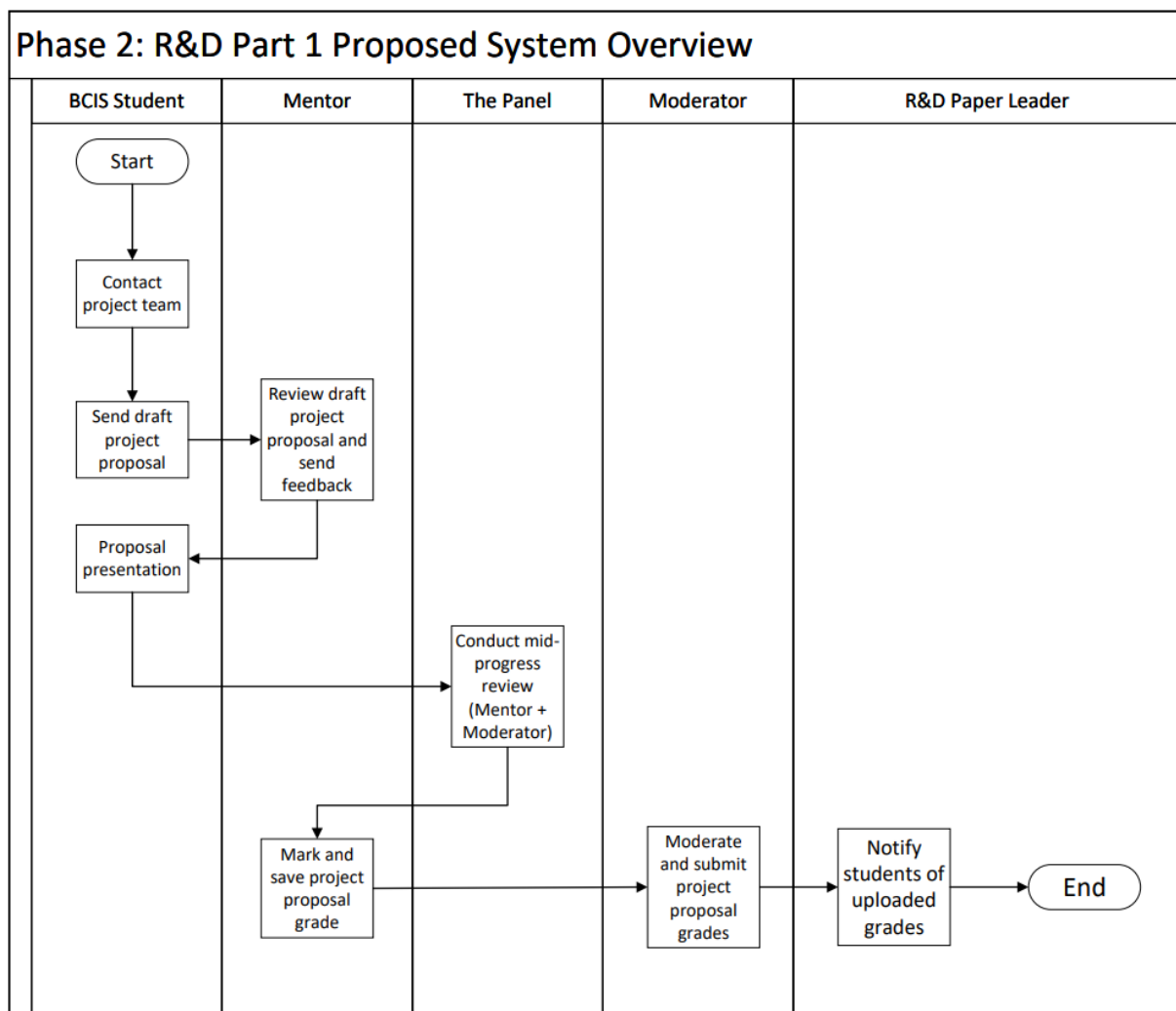


Figure 9 describes the proposed R&D part 1 phase. The processes from the start to “Conduct mid-progress review” remain the same as in the current system. The processes from “Mark and save project proposal grade” to “Notify students of uploaded grades” will be performed on Canvas. By using Canvas it eliminates the issues of marking forms not saving and duplicate teams existing. Canvas also removes the need to publish the grades as Canvas and Arion will be integrated. This means the actor “CMS Administrator” will not be needed as the “R&D Paper Leader” actor can post an announcement on Canvas that will simultaneously email the students of the uploaded grades.

Fig 10. R&D Part 2 Proposed System Overview

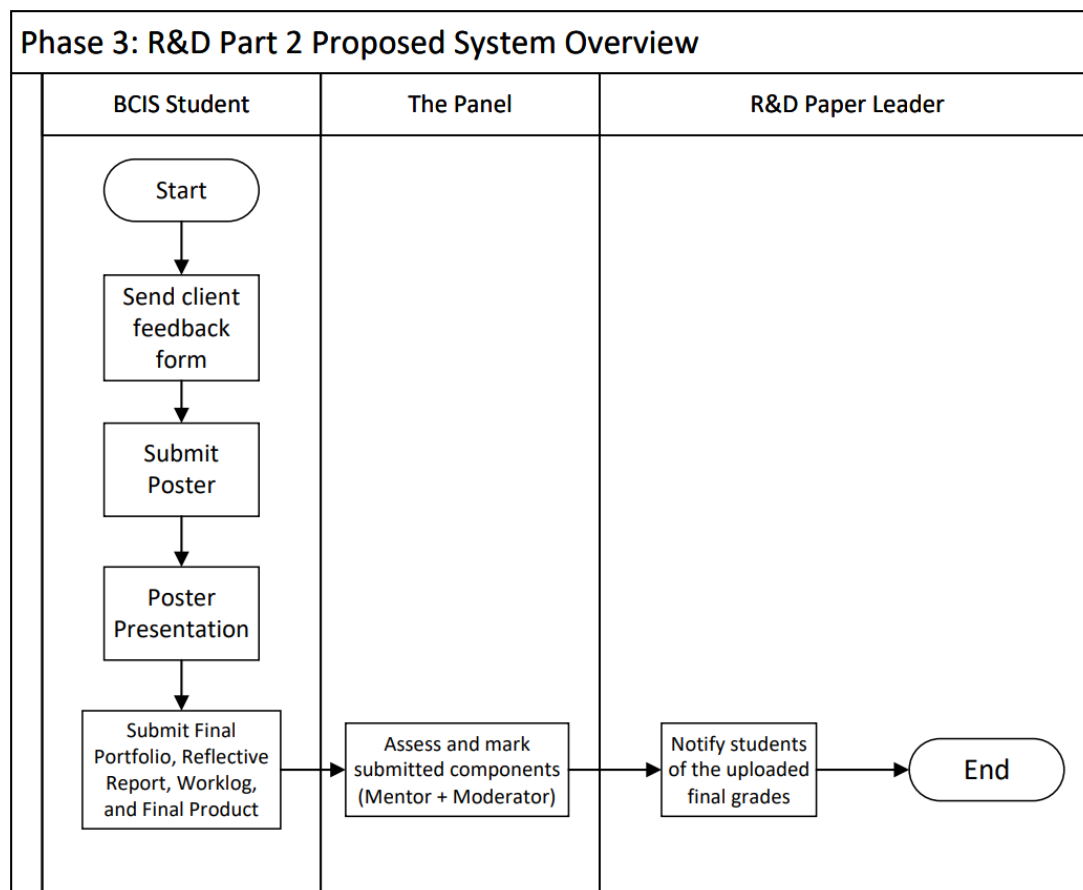


Figure 10 describes the proposed R&D part 2 phase. The processes from the start to “Submit Final Portfolio, Reflective Report, Worklog, and Final Product” remain the same as in the current system. Usage of Canvas nullifies the issues that SharePoint has, it also removes the need for the “CMS Administrator” actor as the processes of publishing will not be needed since Canvas and Arion will be integrated. For the students to receive notification of their uploaded grades the “R&D Paper Leader” actor will post an announcement on Canvas which will simultaneously email the students.

8.3 Summary of Improvements

- Previous issues with grading would be solved as Canvas would be replacing SharePoint in this process
- Previous issues with duplicate projects and teams would be solved as Canvas would be replacing SharePoint in this process
- There will be a dedicated team for support with Canvas
- Canvas has a feature that allows assessors to grade individuals of a group
- Paper leader can upload an announcement of uploaded grades which will simultaneously email the students which also means the CMS admin would not need to notify the students
- Canvas will be integrated with arion which will eliminate the double entry input for the student grades
- Canvas allows the mentors to save the grades using the rubrics tool and the moderator will submit those saved grades after moderation, this eliminates the process of sending the grades from the mentor to the moderator and eliminates the publishing process

8.4 Time and Resource Costs

- Time will need to be spent for the BCIS staff (HOD, mentors, moderators, paper leader, admins, etc.) in learning how to use Canvas
- Time will also need to be spent to produce a guidebook on Canvas
- Mentors, moderators and paper leader will then need to spend time reading the guidebook

8.5 Impacts

8.5.1 Software Impacts

- Processes of SharePoint such as team allocation, team management, and grading will be performed on Canvas instead

8.5.2 Organizational & Operational Impacts

- Blackboard will be replaced by Canvas
- altLab team will need to teach IT department on how to use Canvas
- The team management and allocation will no longer be done on SharePoint instead it will be done on Canvas

8.5.3 Rationale

- Canvas offers more flexibility and functionality with team management
- Creating a more stable environment than SharePoint for mentors, moderators, admins, and paper leader
- Canvas offers rubric that can modified at any time

- Canvas will be integrated with Arion
- Canvas will be supported by IT
- We identified no issues about using SharePoint for the EOI process and project and client management

9. Alternative System

9.1 Description

This system requires the use of Microsoft Excel and Microsoft Forms for the main processes while still using Canvas for a minor role. Canvas' role in this system is to give students access to the EOIs by providing them a link that leads them to a Microsoft Forms page as well as to handle the submissions. Canvas will simultaneously provide a Microsoft Excel sheet along with the EOIs which contains all the projects with their relevant details. Microsoft Excel will also be used to store the results of the student's EOIs as Microsoft Forms has a feature that allows them to export it as an .xlsx file. In addition, Microsoft Excel will be used for the processes of managing and storing the teams which means the team allocation process will still be done manually. Another role Microsoft Excel will be undertaking is the grading, which has been done before the implementation of SharePoint.

9.2 Summary of Improvements

- Excel spreadsheets can be downloaded for offline access
- Use of Microsoft Excel has been previously used for storing projects with their relevant information, marking forms and student EOIs
- Only authorised personnel in the organisation have access to the spreadsheets and forms
- Dedicated IT support
- Forms provides a quick overview of the results through the use of graphs

9.3 Time and Resource Costs

- Time will need to be spent for the BCIS admins and paper leader to learn how to use Microsoft Forms effectively
- Time will need to be spent for the paper leader in learning how to use Canvas

9.4 Impacts

9.4.1 Software Impacts

- The processes of grading, managing and allocating the teams will be performed on Microsoft Excel

9.4.2 Organizational & Operational Impacts

- Blackboard will be replaced by Canvas
- The project list file will need to be uploaded every time when edits are made
- The grading, team management and allocation will no longer be done on SharePoint instead it will be done on Microsoft Excel

9.5 Rationale

- Reduce errors in grading by using Microsoft Excel as it has been used for grading before
- Make SharePoint redundant because the person in charge has left and ever since there have been many issues with it
- Keep using Microsoft Excel for storing the students' EOI information, the projects and their relevant details as we have not identified any issues with it
- Reduce errors in team management by not using SharePoint (duplicate teams, etc.)
- Microsoft Excel and Forms will be supported by IT
- Creating a more stable environment than SharePoint for mentors, moderators, admins, and paper leader

Appendices

Appendix A Stakeholder Register

Appendix B Stakeholder Management Strategy

Appendix C Thematic Synthesis Diagrams

Appendix D As-is System Modelling (full version)

Appendix E To-be System Modelling (full version)

Appendix F Mentor and Student Survey Summary